A Guide to the Data Science Lifecycle

[Last Updated16/01/25]

Table of Contents

Introduction

How to Use this Guide

The aim of this guide is to help you navigate the Data Science Lifecycle (DSLC) framework. Below is a breakdown of how to use each section within the guide:

Section Name	Description	Objective
Data Science Lifecycle Overview	Provides a high-level overview of the DSLC and its aims	Understand the DSLC and its three key stages
Completing the DSLC Checklist	Offers a step-by-step guide to completing the checklist.	Understand how to complete the checklist
Roles and Responsibilities	Defines example roles and responsibilities involved in the DSLC process	Understand the specific duties and expectations of each role
Supporting artefacts and templates	Explains how supplementary artefacts and templates can support the DSLC process	Recommended resources that streamline tasks and promote consistency in documentation
Use Case and Model Inventory Storage	Explains how to manage use cases and model inventory using Confluence.	Understand how to navigate Confluence pages, follow the defined structure, and update or modify entries to maintain relevance
Frequently Asked Questions (FAQ)	Addresses common questions about the DSLC process and its implementation.	Get answers to frequently asked topics.

General Tips:

1. Use the guide as a reference at each stage to stay aligned with DSLC requirements.
2. Ensure team members are aware of roles and responsibilities for completing checklist items.
3. Access all referenced templates, tools, and documentation through the linked SharePoint or Confluence pages.
4. Keep checklists and supporting evidence up to date to streamline sign-off and QA reviews.

Data Science Lifecycle Framework

The Sky Data Science Lifecycle (DSLC) Framework is designed to guide the end-to-end process of delivering a successful Data Science use case—from data exploration, through model development and deployment, to operational monitoring and eventual deprecation.

Figure 1- Data Science Lifecycle Framework

The DSLC Framework is divided into three key stages:

- **Stage 1: Proof of Value (POV)** – *Ensures that use case objectives are aligned with business priorities, establishing a foundation for the meaningful impact. Key outcomes include validating the use case scope, data preparation and analysis activities, and developing solution components required for analytics-based insights and/or models.*
- **Stage 2: Productionisation** –*Translates the POV findings into a robust, scalable, and fully operational solution ready to deliver a meaningful impact. Key outcomes include the successful production deployment, the development of automated data pipelines, and the completion of comprehensive end-to-end testing.*
- **Stage 3: Operationalisation** – *Focusses on continuous monitoring of model performance to ensure meaningful impact is sustained. Key outcomes include establishing a periodic review process and defining next steps based on QA outcomes.*

For a detailed breakdown of each stage, including objectives, outcomes, and best practices, refer to the DSLC Framework.

How to fill the DSLC Checklist

This section outlines the steps required to fill out the DSLC checklist, ensuring traceability, accountability, and quality assurance throughout the lifecycle.

Information Retrieval

- **Objective**: Gather relevant details such as decks, SharePoint links, and Teams channels for the current DSLC stage.
- **Steps**:
    - Review the existing documentation to understand objectives and priorities.
    - Retrieve all necessary materials from relevant stakeholders.

Obtain DSLC Templates

- **Objective**: Access and utilize standardized checklist template completion.
- **Steps**:
    - Download templates from the **SharePoint**
    - Rename the template as followed *'Stagename_usecasename_vx_ddmmyy'*

Link Supporting Evidence

- **Objective**: Provide supporting evidence for each checklist item to ensure transparency and traceability.
- **Steps**:
    - Identify relevant documentation stored in SharePoint or other repositories.
    - Link evidence directly to checklist items for easy reference.
    - Ensure the evidence aligns with the checklist requirements.

Review and Justify Exceptions

- **Objective**: Handle incomplete checklist items through documented justifications.
- **Steps**:
    - Identify items that cannot be completed due to shifting priorities.
    - Document exceptions with a clear rationale and submit for approval.

DSLC Checklist: Step-by-Step

The Excel checklist template is locked to safeguard the structure, with only the sections requiring input left editable. These editable sections must be completed by the responsible or accountable owner for the use case. Following are examples of these section:

Sky Data Science Lifecycle

Stage 01 - POV

Team	*Replace with the relevant information*
Pillar	*Replace with the relevant information*
Checklist Version	*Replace with the relevant information*
Project/Use Case Name	*Replace with the relevant information*
Stage Gate Accountable Owner	*Replace with the relevant information*

Figure 2 - Basic Details Section

1. Navigate to the first tab (names as 'Summary') on the given checklist and at the top of the page begin by completing the essential information at the top of the checklist.

Team: Enter the name of the team responsible for the use case.
Pillar: Specify the relevant pillar or category.
Checklist Version: Write down the version number of the checklist to ensure version control 'vx_ddmmyy'.
Use Case Name: Fill in the name or identifier of the specific use case this checklist addresses.
Stage Gate Accountable Owner: Provide the full name of the person completing the checklist.

Date Completed	DD/MM/YYYY	Date Approved	DD/MM/YYYY
Step 1a - Use Case Shaping			
Definition: Validate the scope of the use case and evaluate the feasibility of the solution. Ensure that key aim, objectives, roles and target deliverables are clearly defined.			
Responsible Owner:			
Accountable Owner:			
Step Progress:	Not Started		

Figure 3 - Dates and Key Roles

2. Proceed to individual steps sheet. The first section within each step provide details for tracking and ownership:

Date Completed: Record the date when the step is filled out.
Date Approved: Enter the date when the step is reviewed and approved by accountable owner.
Responsible Owner: Input the name of the person responsible for the filling out the checklist (usually DS or MLE).
Accountable Owner: Input the name of the individual who will be reviewing the checklist (i.e. DS Lead).
Step Progress: Select the state of the Step using the dropdown.

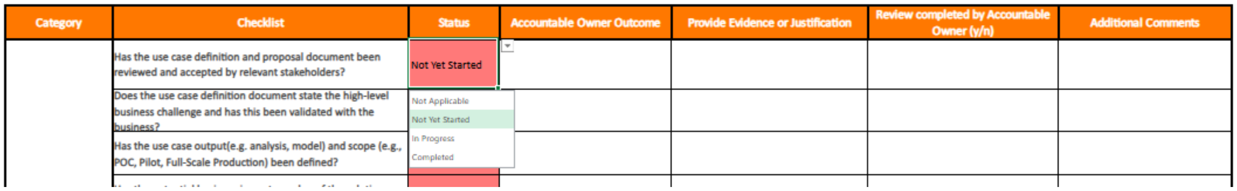


Figure 4 - Dropdown Status

3. For each item in the checklist click the dropdown menu.

- Select the most accurate option for each item using the provided dropdown menu.
- Ensure the status reflects the current state of the item in the step. Refer to the legend in the Summary tab for definition of each status.

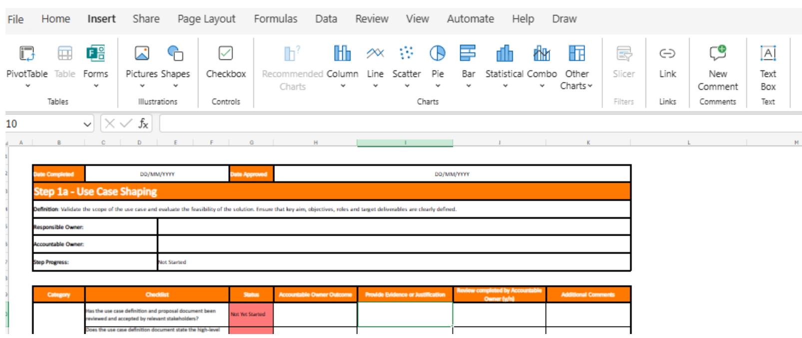


Figure 5 - Evidence or Justification Section

4. For each item ensure that you:

- Provide evidence or justification to support the status of each step.
- Write a concise but clear explanation of what has been done to meet the requirement.

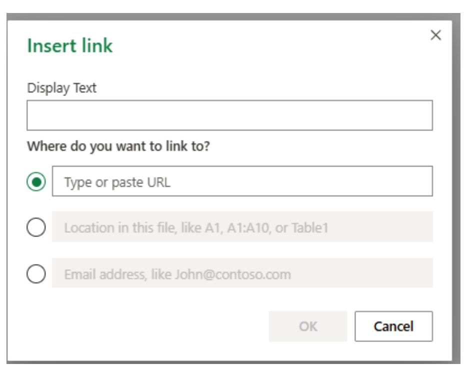


Figure 6 - Linking Evidence in Excel

To attach evidence:

- Insert a hyperlink into the checklist linking to the evidence file or relevant document.
- In Excel, right-click on the cell where the evidence link should be inserted.
- Select Insert Hyperlink and paste the URL or path to the document.
- Label the link clearly (e.g., "Evidence File for Step 1a") so it is easy to understand

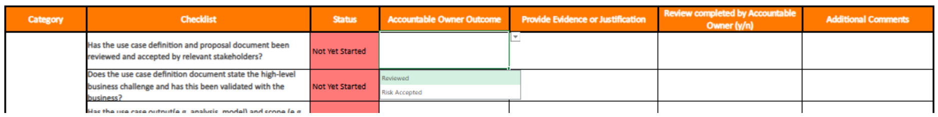


Figure 7 - Reviewing Evidence

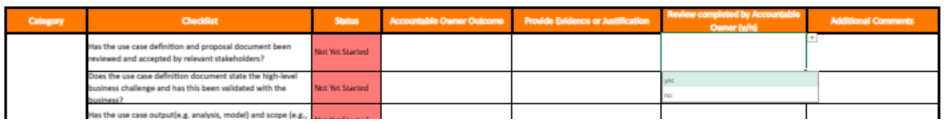


Figure 8 - Final Verification

5. After evidence has been linked for all the items, change the **Step Progress** status to 'Ready to be reviewed' in first section of the sheet and notify the accountable owner: The Accountable owner should:

- Review evidence carefully to ensure it aligns with the checklist requirements.
- Mark each item as 'reviewed' or 'risk accepted' in the **Accountable Owner Outcome** column.
- If no further changes are required for an item and its evidence, mark the **Review Completed by Accountable Owner** as 'yes'
- Repeat this process for all the items in a step

Grouping Patterns for Checklist for Multi-Models Use Cases

The following table provides example mappings for checklist steps when working with use cases involving multiple models. Depending on the use case and model type, the same checklist can be reused for multiple models if the information and processes are similar. Separate steps should be created only for areas where differences exist.

The table below illustrates examples for Stage 1 of where a single checklist step can be used for multiple models or use cases, and where a separate step might be required. Note that this is not an exhaustive list and can be adapted as needed based on the specific situation.

Conditions					**Stage 1 Common Steps**	**Stage 1 Separate Steps**	**Example Use Cases**
Models Use Same Data Source	*Models Have similar Features*	*Model Type is Identical*	*Model Outputs are Similar*	*Use Case Territory*			
✓	✓	✓	✓	✓	All	-	Winback UK
✓	✓	✓	✗	✓	1a, 1b, 1c, 1d,	1e, QA	IDR
✓	✗	✗	✗	✓	1a, 1b	1c, 1d, 1e	Call Forecasting
✗	✗	✗	✗	✓	1a	1b, 1c, 1d, 1e	WHIX
✗	✓	✓	✓	✗	1c, 1d, 1e	1a, 1b, QA	Winback Ireland

Taking an Example for IDR, use case have two model of same type using same data source with different labels. The only different steps in this case will be 1e . Model training, Eval.

If the same use case being used in different territory (such as Winback Ireland) 1a- use case shaping (because customer base, stakeholders, and different regulations) and 1b. data Exploration (because different data source) will be different with a separate QA for outputs of the said territory. However, if there are items that are similar source territory (i.e. Winback Uk), those items can be referenced by linking the source territory's checklist as evidence.

For use cases with substantial number of models (i.e., call forecasting with 90+ models) a grouping of models can created based on the overlapping similarities.

For Ad hoc Analysis Use case, all model development related steps will Be Not Applicable for adhoc use cases so only step 1a, 1b and 1c are applicable and then skipping to QA and Sign off directly.

Roles and Responsibilities

Detailed view of the recommend roles and responsibilities can be found in the SharePoint folder here.

Supporting Artefacts and Templates

Below are the model and use case supplementary artifacts and templates that can be used to support documentation withing various stages of the Data Science Life Cycle (DSLC) process. More detail on each artefact can be found in Supporting Documents on Sharepoint.

These artefacts templates are optional, that means team can keep using any existing templates while making sure they cover all the necessary information highlighted in these templates. These artefacts can help as evidence for the checklist during various stages.

Model Card

Compiling detailed information about the model's provenance, intended use and data governance using a standard template. This can also be used as Model One pager for Business stakeholders.

Template can be found here: Model Card Template

Model Report

Technical details of model primarily for the MLOps and DS to verify expected model behaviour in Stage 2- Productionise. This can be of any format, ideally generated through automation.

Model Report

Use Case One Pager

High-level details of a use case shared amongst wider stakeholders. Template can be found here: Use Case One Pager

Use Case Proposal

Contains a detailed plan to justify the development or prioritization of a specific use case. Template can be found here: Use Case Proposal

Sign off and Deployment Review

Template to facilitate the Signoffs meeting at end of each stage. Template can be found here: Sign off and Deployment Review Meeting Templates

Model and Business Monitoring

Guidelines for tracking, managing live model with measurable business goals in mind. More details can be found here: Data Science Lifecycle - Model and Business Monitoring .pdf

When to create these artefacts

Artifact	Completed by
Model Card	End of Stage 1
Model Report	Before deployment in Stage 2
Use Case One Pager	End of Stage 1
Use Case Proposal	Before Starting Stage 1
Sign-off and Deployment Review	Before Sign-off meetings in each Stage
Confluence	End of Stage 1

Confluence for Use Case and Model Inventory

Confluence Structure

This section explains how to store and manage use case details and model inventory efficiently within Confluence.

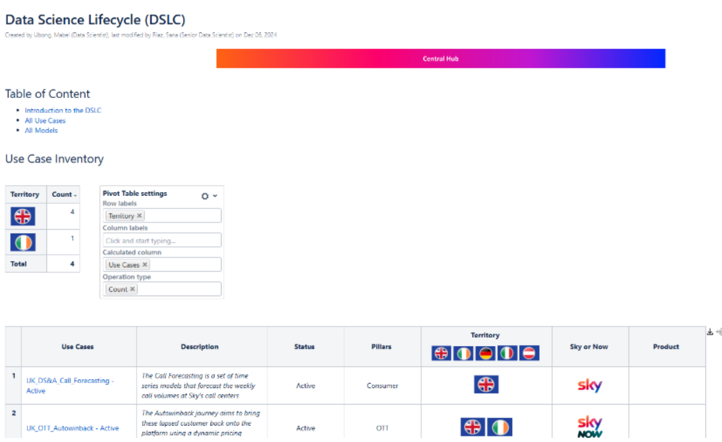


Figure 9- DSLC Confluence Landing Page

This serves as the central hub for all Use cases migrated to DSLC framework.

Figure 10 - Schematic Structure of the DSLC Confluence

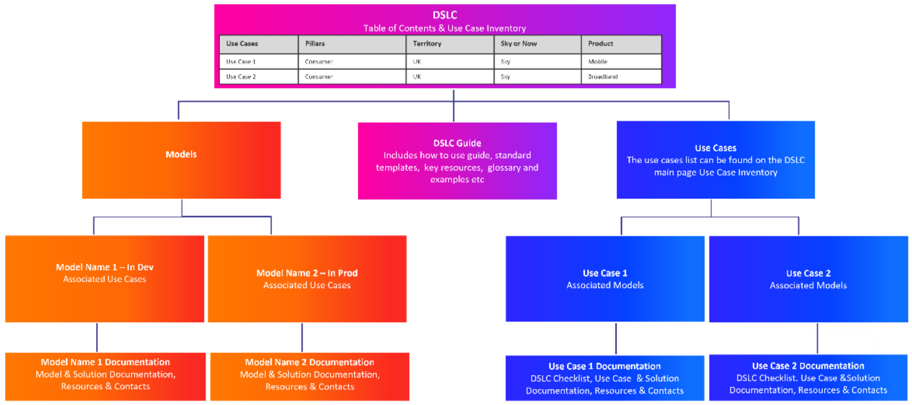


Figure 11 - Screenshot of Confluence Structure as of 17/12/24

Use case and Model Pages Templates

An example structure is added for the use case template and model template in the respective folders in the confluence. These contain sections where links to the relevant resources need to be added and can be adapted to new use cases, as necessary.

When filling the confluence pages, any stage 2 related information is the responsibility of Machine Learning Engineers, whilst the rest should be completed and maintained by the Data Scientist.

Use case and Model Pages Naming Convention

The following naming convention is used for the use case pages:

Use Case Main Page: Territory_Team_UseCase_Name – UseCase_Status

Use Case Documentation Page: Territory_Team_UseCase_Name Documentation

For example, for IDR this would be *‘UK_DS&A_IDR – Active’* for main page and ‘*UK_DS&A_IDR – Documentation’* for documentation page. The status of the use case can be *‘Active’* or *‘Inactive’*.

Similar naming convention is followed by model pages, with trailing ‘_model’ at the end of the name to differentiate between use case and model pages. I.e.

Model Main Page: Territory_Team_Model_Name_Model – Model_Status

Model Documentation Page: Territory_Team_ Model_Name_Model Documentation

The status for the model can be ‘*In Dev*’, ‘*In Prod*’, *‘Live’* or *‘Decommissioned’*.

Managing Confluence Pages

Following are the details on everything you need to complete in confluence When adding a new use case on confluence:

1. Create a use case page for your use case within ‘All Use Cases’ level and the relevant documentation page. Please use the templates provided.

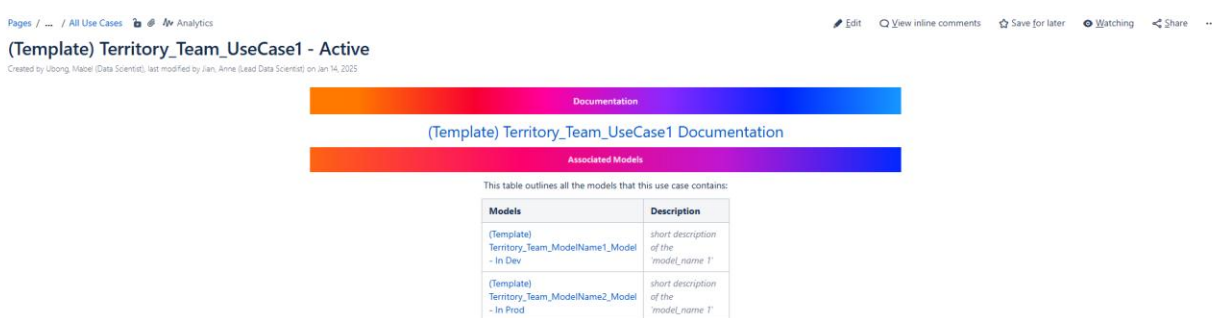


Figure 122 - Screenshot of Confluence Use Case Page 1 Template

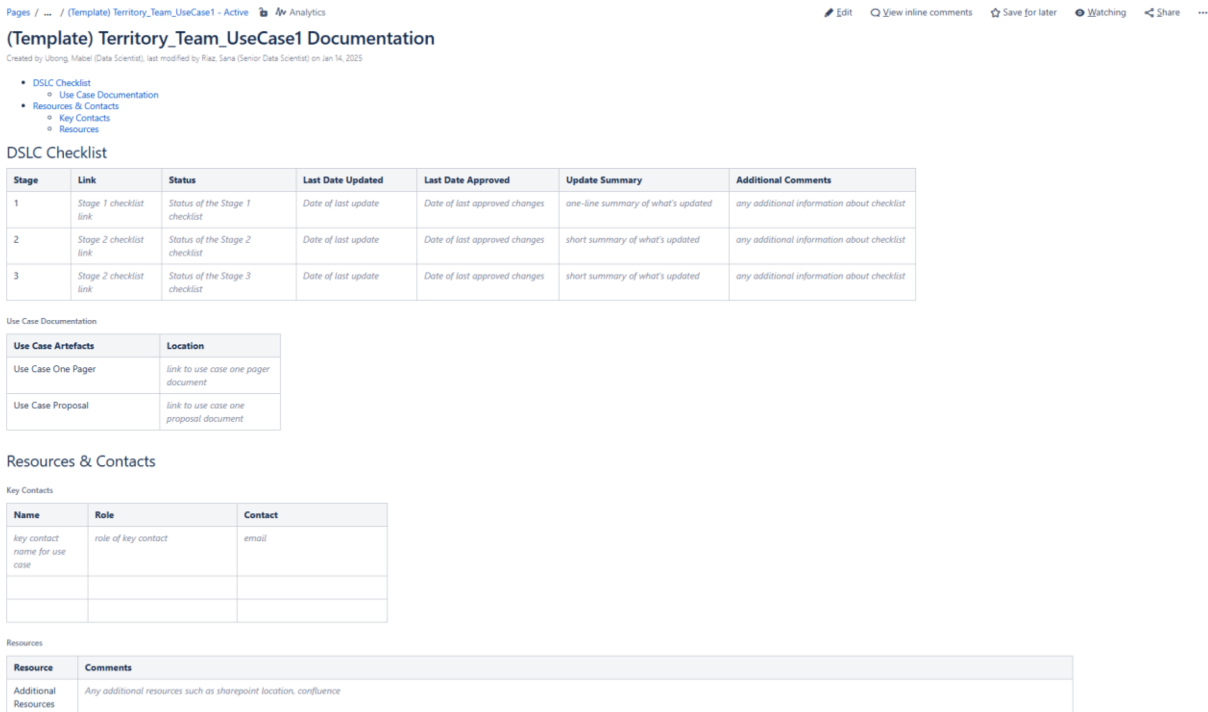


Figure 13 - Screenshot of Confluence Use Case Page 2 Template

2. Create the models page within '[All Models](#)' level and then link the model to use cases in the relevant section. Please use the templates provided.

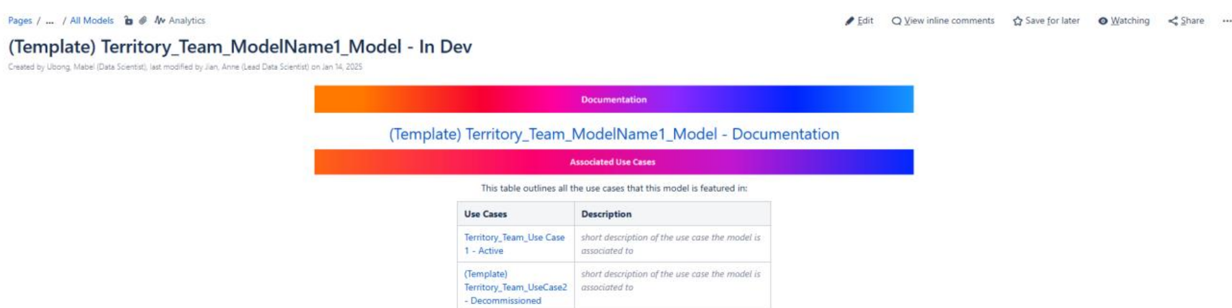
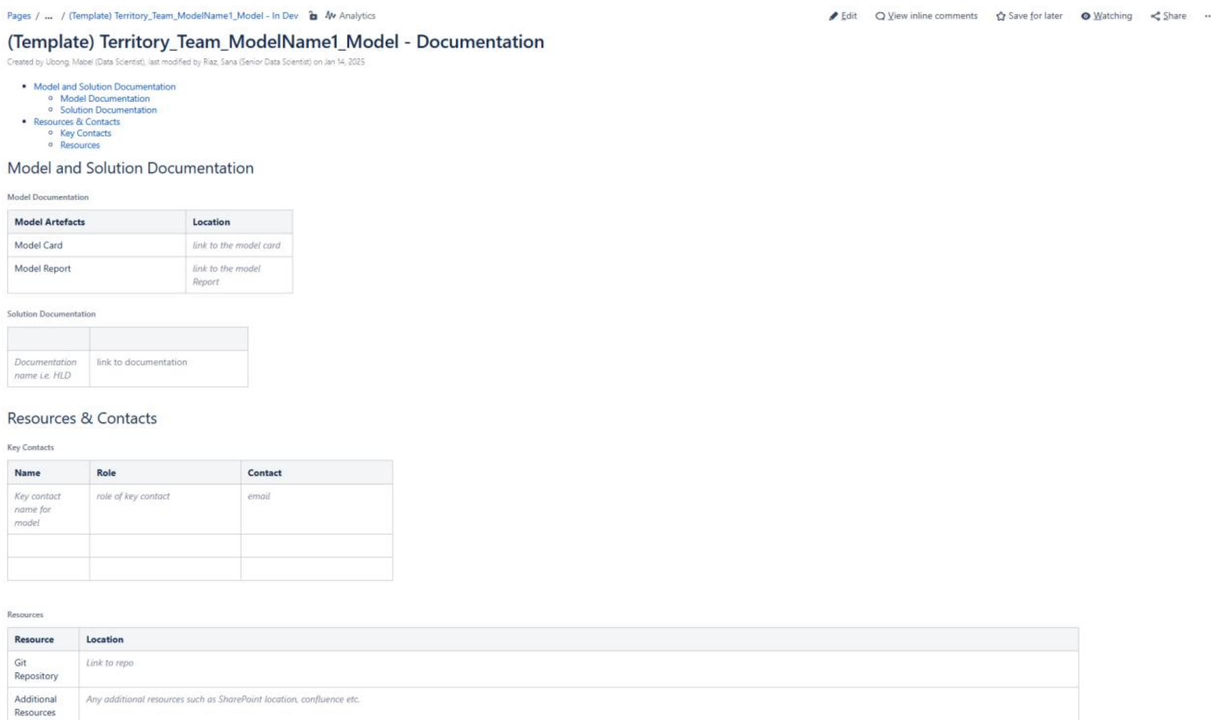


Figure 134 - Screenshot of Confluence Model Page 1 Template



Pages / ... / (Template) Territory_Team_ModelName1_Model - In Dev Analytics

Edit View inline comments Save for later Watching Share ...

(Template) Territory_Team_ModelName1_Model - Documentation

Created by Ubong, Mabel (Data Scientist), last modified by Riaz, Sana (Senior Data Scientist) on Jan 14, 2025

- Model and Solution Documentation
  - Model Documentation
  - Solution Documentation
- Resources & Contacts
  - Key Contacts
  - Resources

Model and Solution Documentation

Model Documentation

Model Artefacts	Location
Model Card	*link to the model card*
Model Report	*link to the model Report*

Solution Documentation

Documentation name i.e. HLD	link to documentation

Resources & Contacts

Key Contacts

Name	Role	Contact
Key contact name for model	*role of key contact*	*email*

Resources

Resource	Location
Git Repository	*Link to repo*
Additional Resources	*Any additional resources such as SharePoint location, confluence etc.*

Figure 145 - Screenshot of Confluence Model Page 2 Template

3. Add in a line for the new use case on the landing page (Data Science Lifecycle (DSLC) - Data Technology & Analytics - DevEx DocHub) for use case inventory, see the red box below on Figure 16.

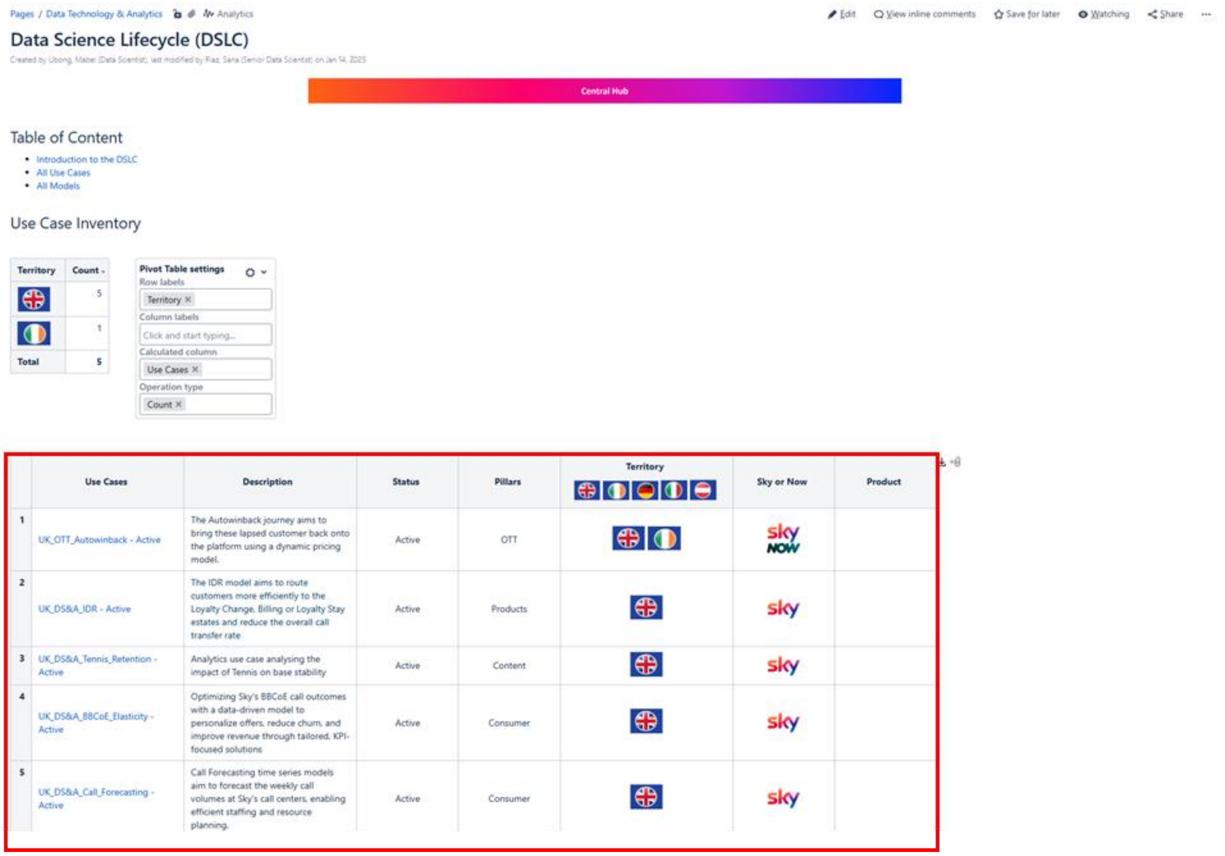


Figure 156 - Screenshot of Confluence Use Case Inventory

Create a new page (for models and use case)

- Use the Search Bar or Sidebar Navigation to locate the space where you want to create the new page.
- Ensure you have the necessary permissions to add content to this space.
- Click the Create button, usually in the top navigation bar or sidebar of the Confluence interface.
- Select the type of page you want to create (e.g., blank page, pre-defined template, etc.). If your space includes pre-defined templates (e.g., for use cases or model inventories), choose the appropriate template to maintain consistency.

Adding Links

- Navigate to the page you would like to edit
- Edit the page by clicking the "Edit" button at the top right.
- Highlight the text you want to turn into a link (e.g., "Step 1").

- Click the link icon in the toolbar or press Ctrl+K.
- Enter the URL or select a Confluence page to link to.
- Click "Insert" to add the link.

Adding Dates

- Edit the page by clicking the "Edit" button.
- Click on the cell under the "Last Date Updated" or "Last Date Approved" column.
- Click the calendar icon in the toolbar.
- Select the date from the calendar.
- Click "Insert" to add the date.

Adding Files

- Edit the page by clicking the "Edit" button.
- Click on the cell under the "File" column.
- Click the paperclip icon in the toolbar to attach a file.
- Upload the file from your computer or select an existing file from Confluence.
- Click "Insert" to add the file.

Adding Key Contacts

- Navigate to the Resources & Contacts section.
- Edit the page by clicking the "Edit" button.
- Add the contact information (e.g., name, title, email) in the appropriate section.
- Highlight the email address and click the link icon in the toolbar or press Ctrl+K.
- Enter the email address in the link field (e.g., mailto:firstname.surname@sky.uk).
- Click "Insert" to add the email link.